



2009

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start a new booklet for each question

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value

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Total marks – 84

Attempt Questions 1–7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet		Marks
(a)	The polynomial $P(x) = x^3 - 2x + 7$ is divided by $x - 5$. What is the remainder?	1
(b)	Differentiate $y = \tan^{-1}\left(\frac{x}{3}\right)$ with respect to x .	2
(c)	Evaluate $\int_0^{\frac{3}{4}} \frac{1}{\sqrt{9-4x^2}} dx$	3
(d)	If $x = 8\cos\theta, y = 5\sin\theta$ find $\frac{dy}{dx}$ when $\theta = \frac{\pi}{4}$.	2
(e)	(i) Graph $y = 1 + \frac{3}{x-2}$ showing all important features such as intercepts and asymptotes	2
	(ii) Hence solve $\frac{3}{x-2} > -1$	2

End of Question 1

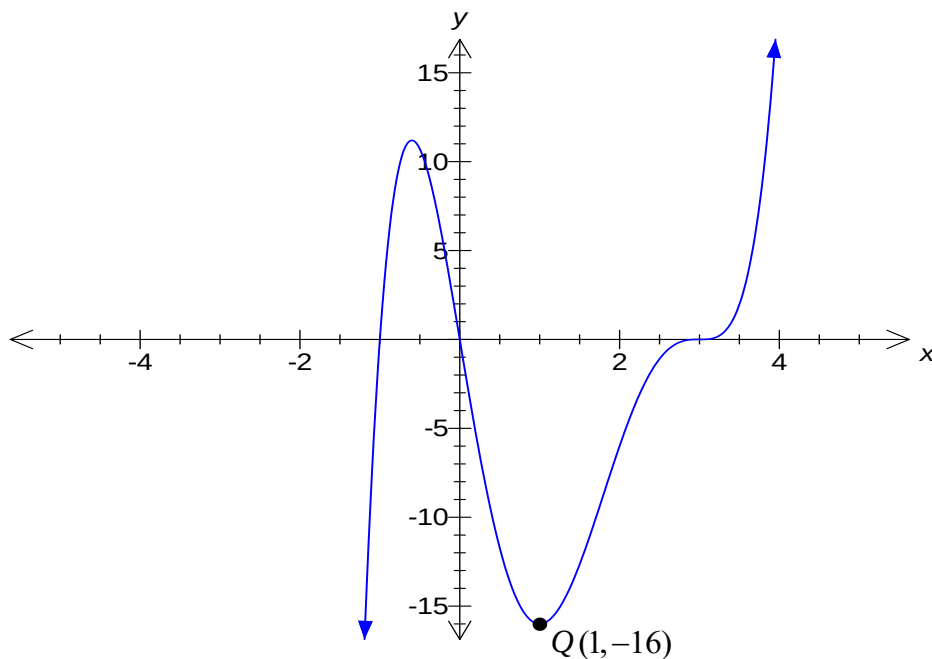
Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Find $\int \sin^2 4x \, dx$

2

(b)



- (i) The above graph has equation $y = ax(x-b)(x+c)^d$.
Write down values for a , b , c and d .

3

- (ii) The approximate root was chosen as $x = 1$. Explain why this choice will not give a better approximation to the root using Newton's Method.

1

- (c) The function $f(x) = \sin x - \log_e x$ has a zero near $x = 2.5$.
Use one application of Newton's method to obtain another approximation to this zero.
Give your answer correct to two decimal places.

2

- (d) (i) How many numbers greater than 50000 can be formed with the digits 2, 3, 5, 7, 9 if no digit repeated?

2

- (ii) What is the probability that if the numbers are selected at random and laid next to each other, that the number formed is greater than 50000 but less than 90000?

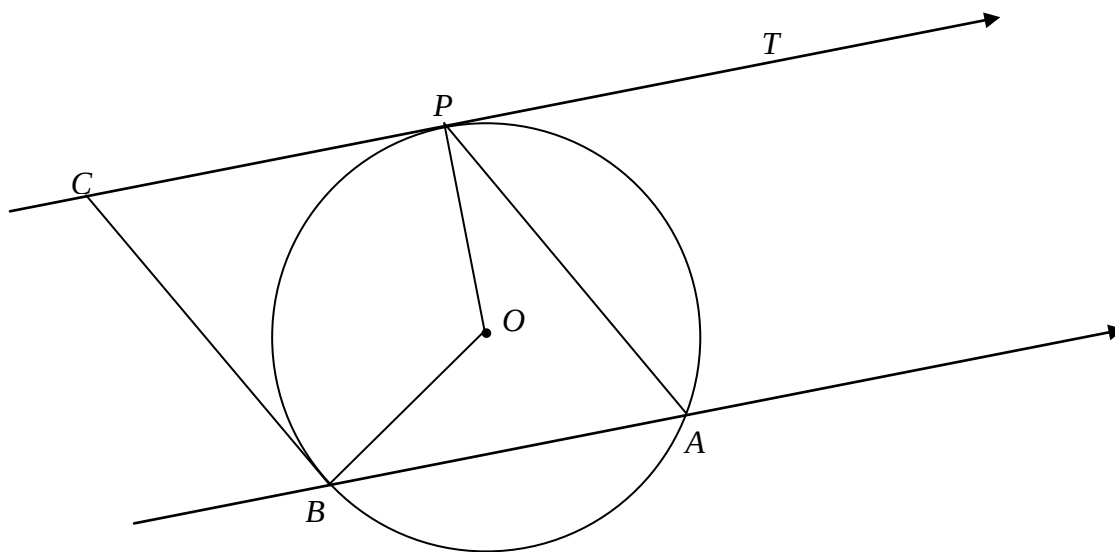
2

End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find, as an integer, the coefficient of x^3 in the expansion of $(2x-3)^9$. 2
- (b) The polynomial $x^3 - 15x^2 + 71x - 105 = 0$ had roots α, β, γ which form an arithmetic sequence with a common difference of 2. Find the value of the three roots. 3
- (c) CT is a tangent to the circle, centre O , touching at P . Quadrilateral $PABC$ is a rhombus and CT is parallel to AB .



- (i) Let $\angle TPA = x$ and prove that $\angle POB = 2x$ 2
- (ii) Find the value of x such that $POBC$ is a cyclic quadrilateral. You must support your answer with geometrical reasons. 2
- (d) In Poker 5 cards are dealt to each player. The deck has 4 suits with 13 cards in each suit. For the following questions leave your answer in unsimplified form.
- (i) How many different hands are possible? 1
- (ii) Find the probability of a “flush” (all five cards are from the same suit). 2

End of Question 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Solve $\sin x + \cos x = 1$, using “t” method, in the domain $0 \leq x \leq \pi$. **3**
- (b) By using the substitution $u = \tan x$, or otherwise, find $\int \frac{\sec^2 x}{\sqrt{1+2 \tan x}} dx$. **3**
- (c) (i) Show that the equations of the tangent and normal to the parabola $x^2 = 4ay$, at the point $P(2ap, ap^2)$ are: **2**
- Tangent: $y = px - ap^2$
Normal: $yp = -x + 2ap + ap^3$
- (ii) T and G are the points where the tangent and normal meet the axis of the parabola. Find the coordinates of these points. **2**
- (iii) If S is the focus of the parabola, explain why P , T and G all lie on the circle with centre S . **2**

End of Question 4

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Prove by mathematical induction that, for all integers $n \geq 1$

4

$$2 + 2^3 + 2^5 + \dots + 2^{2n-1} = \frac{2(2^{2n} - 1)}{3}.$$

- (b) If $f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$

- (i) Show that $f'(x) = 0$

2

- (ii) State the domain of $f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$

1

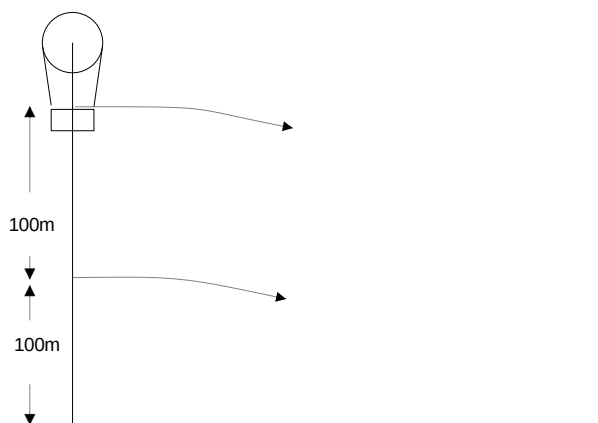
- (iii) Sketch $f(x)$ in its domain

2

- (c) A balloon rises vertically from level ground. Two projectiles are fired horizontally in the same direction from the balloon at a velocity of 80ms^{-1} . The first is fired at a point 100 m from the ground and the second when it has risen a further 100 m from the ground.

3

How far apart will the projectiles hit the ground (give your answer to the nearest metre)?
(Use $g = 10\text{ms}^{-2}$)

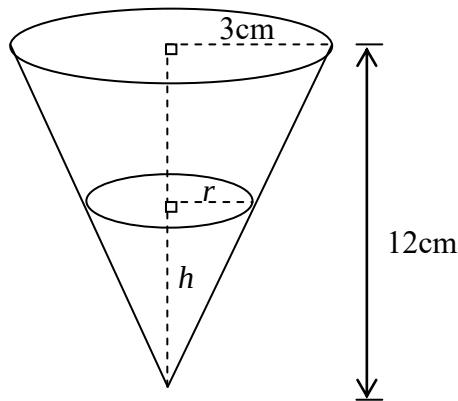


End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) The diagram shows a conical drinking cup of height 12cm and radius 3cm. The cup is being filled with water at the rate of 2 cm^3 per second. The height of the water at time t seconds is h cm and the radius of the water's surface is r cm



- (i) Show that $r = \frac{1}{4}h$ 1
- (ii) Find the rate at which the height is increasing when the height of the water is 8 cm. 3
 (Volume of a cone $= \frac{1}{3}\pi r^2 h$)
- (b) A household iron is cooling in a room of constant temperature 23°C . At time t minutes its temperature T decreases according to the equation $\frac{dT}{dt} = -k(T - 23)$ where k is a positive constant. The initial temperature of the iron is 90°C and it cools to 70°C after 10 minutes.
- (i) Verify that $T = 23 + Ae^{-kt}$ is a solution of this equation where A is a constant. 1
- (ii) Find the exact values of A and k . 2
- (iii) How long will it take for the iron to cool to a temperature of 40° ? 2
 Give your answer to the nearest minute.
- (c) If $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$ prove $\sum_{r=1}^n r {}^nC_r = n \cdot 2^{n-1}$ 3

End of Question 6

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(Question 7 continues on page 10)

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) A projectile is fired from the origin O with velocity V and with angle of elevation θ ,
Where $\theta \neq \frac{\pi}{2}$. You may assume that:

$$x = Vt \cos \theta \text{ and } y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

where x and y are the horizontal and vertical displacements of the projectile in metres from O at time t seconds after firing.

- (i) Show that the equation of flight (trajectory) of the projectile can be written as:

3

:

$$y = x \tan \theta - \frac{x^2}{4h}(1 + \tan^2 \theta) \quad \text{where } h = \frac{V^2}{2g}$$

- (ii) Show that the point (X, Y) , where $X \neq 0$, can be hit by firing the projectile at two different angles θ_1 and θ_2 provided:

3

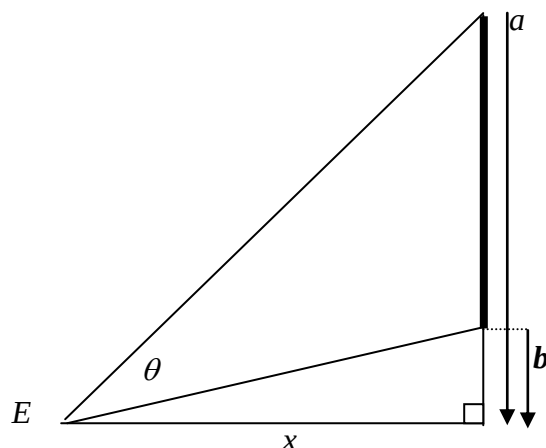
$$X^2 < 4h(h - Y)$$

Question 7 continued on page 11

Question 7 (continued)

Marks

- (b) An observer's eye E looks up at a large painting on a vertical wall. The top of the painting is a metres above the level of E and the bottom of the painting is b metres above the level of E . θ is the angle subtended at the observer's eye by the top and bottom of the painting. E is x metres from the wall. The observer can move backwards and forwards changing x to find the position of best view when θ is a maximum.



- (i) Explain why $\theta = \tan^{-1} \frac{a}{x} - \tan^{-1} \frac{b}{x}$ 1
- (ii) Show that $\frac{d\theta}{dx} = \frac{(a-b)(ab-x^2)}{(a^2+x^2)(b^2+x^2)}$ 3
- (iii) If $a = 3b$, find the maximum possible value for θ . 2

End of Test

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2}) \quad \text{NOTE: } \ln x = \log_e x, \quad x > 0$$



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TRIAL EXAMINATION**

Mathematics Extension 1 (SOLUTIONS)

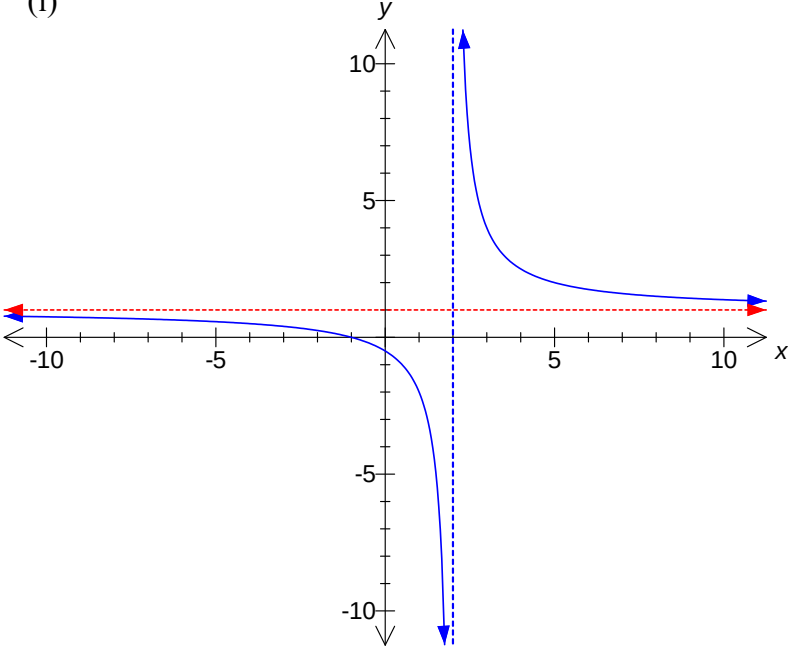
General Instructions

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Question	Criteria	Marks	Bands
1(a)	$P(x) = x^3 - 2x + 7$ Remainder is $P(5) = 5^3 - 2 \times 5 + 7 = 122$	1	
1(b)	$y = \tan^{-1}\left(\frac{x}{3}\right)$ $\frac{dy}{dx} = \frac{1}{1 + \left(\frac{x}{3}\right)^2} \times \frac{1}{3} \quad \checkmark$ $= \frac{3}{9 + x^2} \quad \checkmark$	2	
1(c)	$\int_0^{\frac{3}{4}} \frac{1}{\sqrt{9 - 4x^2}} dx = \int_0^{\frac{3}{4}} \frac{1}{\sqrt{4\left(\frac{9}{4} - x^2\right)}} dx$ $= \int_0^{\frac{3}{4}} \frac{1}{2\sqrt{\left(\frac{9}{4} - x^2\right)}} dx \quad \checkmark$ $= \frac{1}{2} \left[\sin^{-1}\left(\frac{2x}{3}\right) \right]_0^{\frac{3}{4}} \quad \checkmark$ $= \frac{1}{2} \left[\sin^{-1}\left(\frac{2 \times \frac{3}{4}}{3}\right) - \sin^{-1}(0) \right]$ $= \frac{1}{2} \sin^{-1} \frac{1}{2}$ $= \frac{1}{2} \times \frac{\pi}{6} \quad \checkmark$ $= \frac{\pi}{12}$	3	
1(d)	$\left. \begin{aligned} \frac{dx}{d\theta} &= -8 \sin \theta \\ \frac{dy}{d\theta} &= 5 \cos \theta \end{aligned} \right\} \quad \checkmark$ when $\theta = \frac{\pi}{4}$, $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx} = 5 \cancel{\cos \frac{\pi}{4}} \times \frac{1}{\cancel{-8 \sin \frac{\pi}{4}}}$ $= \frac{-5}{8} \quad \checkmark$	2	

1(e)	(i)  ✓ hyperbola and intercepts ✓ asymptotes (ii) $\frac{3}{x-2} > -1$ is when $y = 0$ in equation $y = 1 + \frac{3}{x-2}$ That is, from the above graph where $x < -1$, $x > 2$	2	
		2	

Question	Criteria	Marks	Bands
2(a)	$\cos 8x = 1 - 2 \sin^2 4x$ $\sin^2 4x = \frac{1}{2}(1 - \cos 8x)$ $\int \sin^2 4x \, dx = \frac{1}{2} \int (1 - \cos 8x) dx \quad \checkmark$ $= \frac{x}{2} - \frac{\sin 8x}{16} + c \quad \checkmark$ $= \frac{8x - \sin 8x}{16} + c$	2	

2(b)(i)	$b = -1$ ✓ $\left. \begin{array}{l} c = -3 \\ d = 3 \end{array} \right\}$ ✓ when $x = 1, y = -16$ $-16 = a(1+1)(1-3)^3$ $a = 1$ ✓	3	
2(b)(ii)	A tangent to the curve at $x = 1$ will be horizontal and will not therefore cut through the x-axis again to produce a new approx. to the root.	1	
2(c)(i)	$f(x) = \sin x - \log_e x, f'(x) = \cos x - \frac{1}{x}$ let the new approx. be x_1 $x_1 = 2.5 - \frac{f(2.5)}{f'(2.5)}$ ✓ $= 2.5 - \frac{\sin 2.5 - \log_e 2.5}{\cos 2.5 - \frac{1}{2.5}}$ $= 2.24$ ✓	2	
2(d)(i)	$\boxed{3} \times \underbrace{\boxed{4} \times \boxed{3} \times \boxed{2} \times \boxed{1}}_{\text{remainder} \checkmark} = 72$ 5, 7, 9 ✓	2	
2(d)(ii)	$\frac{2 \times 4! (\checkmark)}{5!} = \frac{48}{120}$ $= \frac{2}{5}$ ✓	2	

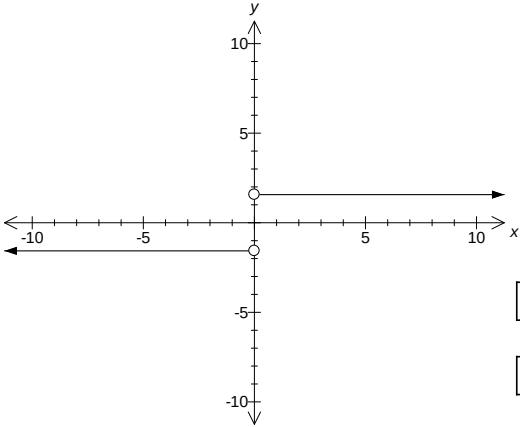
Question	Criteria	Marks	Bands
3(a)	${}^9C_6 (2x)^3 (-3)^6 = 489888x^3$ ✓✓ $\therefore 489888$	2	
3(b)	let the roots be $\alpha - 2, \alpha, \alpha + 2$ ✓ sum of the roots : $\alpha - 2 + \alpha + \alpha + 2 = 15$ ✓ $3\alpha = 15$ $\alpha = 5$ $\therefore$ roots are: 3, 5, 7 ✓	3	
3(c)(i)	$\angle PAB = x$ (alternate $\angle$'s; $CT \parallel BA$) ✓ $\angle POB = 2x$ (angle at the centre is twice the angle on the circumference) ✓	2 Nb -1 for both reasons	

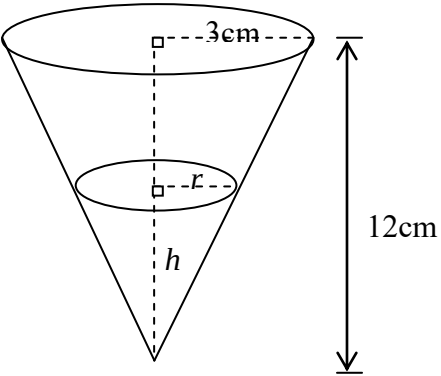
		wrong	
3(c)(ii)	$\angle PCB = x$ (opposite angles of a rhombus are equal)✓ $x + 2x = 180$ (opposite angles of a cyclic quadrilateral)✓ add to 180° $\therefore x = 60^\circ$	2	
3(d)(i)	${}^{52}C_5$ ✓	1	
3(d)(ii)	number of possible = ${}^{13}C_5 \times 4 = 5148$ ✓ $P(\text{flush}) = \frac{5148}{2598960} = \frac{33}{16660}$ ✓	2	

Question	Criteria	Marks	Bands
4(a)	$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1$ ✓ $2t + 1 - t^2 = 1 + t^2$ $2t^2 - 2t = 0$ $2t(t-1) = 0$ $t = 0, 1$ ✓ $\tan \frac{x}{2} = 0$ $\tan \frac{x}{2} = 1$ $\frac{x}{2} = 0$ $\frac{x}{2} = \frac{\pi}{4}$ $x = 0, \frac{\pi}{2}$ ✓ Test $x = \pi$; $\sin \pi + \cos \pi = -1$ $\therefore x \neq \pi$	3	
4(b)	$\int \frac{\sec^2 x}{\sqrt{1+2\tan x}} dx$ let $u = \tan x$ $du = \sec^2 x dx$ $\int \frac{du}{\sqrt{1+2u}}$ ✓ $= \int (1+2u)^{-\frac{1}{2}} du$ $= \frac{2(1+2u)^{\frac{1}{2}}}{2} du$ ✓ $= \sqrt{1+2\tan x} + c$ ✓	3	

4(c)(i)	$y = \frac{x^2}{4a} \quad \frac{dy}{dx} = \frac{2x}{4a};$ at $(2ap, ap^2) \quad \frac{dy}{dx} = p$ $y - ap^2 = p(x - 2ap)$ tangent: $y = px - ap^2 \quad \checkmark$ $y - ap^2 = -\frac{1}{p}(x - 2ap)$ $yp - ap^3 = -x + 2ap$ normal: $yp = 2ap + ap^3 - x \quad \checkmark$	2	
4(c)(ii)	$(0, -ap^2) \quad \checkmark$ $(0, 2a + ap^2) \quad \checkmark$	2	
4(c)(iii)	GT forms the diameter of a circle through P since $\angle GPT = 90^\circ \quad \checkmark$ Midpoint of GT $x = 0 \quad y = \frac{2a + ap^2 - ap^2}{2} = a$ $\therefore$ centre of circle is $S(0, a) \quad \checkmark$	2	

Question	Criteria	Marks
5(a)	$2 + 2^3 + 2^5 + \dots + 2^{2n-1} = \frac{2(2^{2n} - 1)}{3}$ Step1: Prove true for $n = 1$ $\begin{aligned} LHS &= 2^{2(1)-1} \\ &= 2 \end{aligned} \qquad \begin{aligned} RHS &= \frac{2(2^{2(1)} - 1)}{3} \\ &= \frac{2(3)}{3} \\ &= 2 \end{aligned}$ $\therefore LHS = RHS$ true for $n = 1$ ☒ proving true for $n = 1$ Step 2: assume true for $n = k$ $ie: 2 + 2^3 + 2^5 + \dots + 2^{2k-1} = \frac{2(2^{2k} - 1)}{3}$ Step3: Prove true for $n = k + 1$ $RTP: 2 + 2^3 + 2^5 + \dots + 2^{2(k+1)-1} = \frac{2(2^{2(k+1)} - 1)}{3}$ $ie: 2 + 2^3 + 2^5 + \dots + 2^{2k+1} = \frac{2(2^{2k+2} - 1)}{3}$ $\begin{aligned} LHS: & 2 + 2^3 + 2^5 + \dots + 2^{2k-1} + 2^{2(k+1)-1} \\ &= 2 + 2^3 + 2^5 + \dots + 2^{2k-1} + 2^{2k+1} \\ &= \frac{2(2^{2k} - 1)}{3} + 2^{2k+1} \quad (\text{using step 2}) \\ &= \frac{2^{2k+1} - 2 + 3 \cdot 2^{2k+1}}{3} \\ &= \frac{2^{2k+1}(1 + 3) - 2}{3} \\ &= \frac{2^{2k+1}(4) - 2}{3} \\ &= \frac{2^{2k+1}(2^2) - 2}{3} \\ &= \frac{2^{2k+3} - 2}{3} \\ &= \frac{2(2^{2k+2} - 1)}{3} \end{aligned}$ ☒ sub step 2 step 3 ☒ proving RTP statement $\therefore$ true for $n = k + 1$ step 4: if it is true $n = 1$ and $n = 1 + 2 = 2$, and it is true for $n = k$ and $n = k + 1$ then it is true for integers $n \geq 1$ ☒	4

5(b)(i)	$f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$ $f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\left(\frac{1}{x}\right)^2} \times \frac{-1}{x^2} \quad \checkmark \text{ differentiation}$ $= \frac{1}{1+x^2} + \frac{1}{\frac{x^2+1}{x^2}} \times \frac{-1}{x^2}$ $= \frac{1}{1+x^2} + \frac{x^2}{x^2+1} \times \frac{-1}{x^2}$ $= \frac{1}{1+x^2} + \frac{-1}{x^2+1} \quad \checkmark \text{ simplification}$ $= 0$	2
5(b)(ii)	all real x ($x \in \mathbb{R}$) except $x = 0$	1
5(b)(iii)	 $\checkmark \quad y = \frac{\pi}{2} \quad x > 0 \text{ and } y = \frac{-\pi}{2} \quad x < 0$ $\checkmark \quad x \neq 0$	2
5(c)	From 100m : $x = 80t \cos(0) \quad y = -5t^2 + 80t \sin(0) + 100 \Rightarrow y = -5\left(\frac{x}{80}\right)^2 + 100 \quad \dots\dots(1)$ $\therefore x = 80t \quad y = -5t^2 + 100$ From 200m : $x = 80t \cos(0) \quad y = -5t^2 + 80t \sin(0) + 200 \Rightarrow y = -5\left(\frac{x}{80}\right)^2 + 200 \quad \dots\dots(2)$ $\therefore x = 80t \quad y = -5t^2 + 200$ Hits ground when $y = 0$ $(1) \quad -5\left(\frac{x}{80}\right)^2 + 100 = 0 \Rightarrow -5\left(\frac{x}{80}\right)^2 = -100 \quad \therefore 358 \text{ metres} \quad \checkmark$ $(2) \quad -5\left(\frac{x}{80}\right)^2 + 200 = 0 \Rightarrow -5\left(\frac{x}{80}\right)^2 = -200 \quad \therefore 506 \text{ metres} \quad \checkmark$ $\therefore$ difference in range is $506 - 358 = 148$ metres.	3

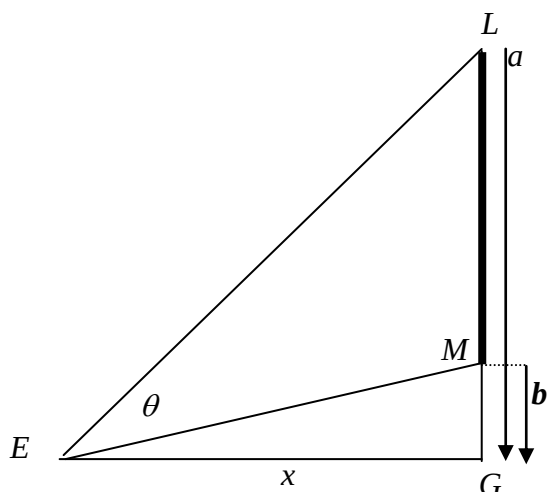
Question	Criteria	Marks
6(a)(i)	 $\frac{h}{r} = \frac{12}{3} \quad \therefore r = \frac{h}{4} \quad \checkmark$	1
6(a)(ii)	$V = \frac{1}{3}\pi r^2 h \text{ and } r = \frac{h}{4} \quad \therefore V = \frac{1}{3}\pi \left(\frac{h}{4}\right)^2 h \Rightarrow V = \frac{\pi h^3}{48}$ $\frac{dV}{dh} = \frac{3\pi h^2}{48} = \frac{\pi h^2}{16} \quad \checkmark$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $\frac{dh}{dt} = \frac{16}{\pi h^2} \times 2 \quad \checkmark$ $\frac{dh}{dt} = \frac{32}{\pi h^2} \quad \text{since } h = 8$ $\frac{dh}{dt} = \frac{1}{2\pi} \text{ cm/s} \quad \checkmark$	3
6(b)(i)	$T = 23 + Ae^{-kt} \Rightarrow Ae^{-kt} = T - 23$ $\frac{dT}{dt} = -kAe^{-kt}$ $= -k(T - 23) \quad \checkmark$	1
6(b)(ii)	$t = 0 \text{ and } T = 90 \quad \therefore 90 = 23 + Ae^{-k(0)} \Rightarrow A = 67 \quad \checkmark$ $t = 10 \text{ and } T = 70 \quad \therefore 70 = 23 + 67e^{-k(10)} \Rightarrow k = -\frac{1}{10} \ln\left(\frac{47}{67}\right) \quad \checkmark$	2
6(b)(iii)	$T = 23 + 67e^{-kt} \text{ where } k = -\frac{1}{10} \ln\left(\frac{47}{67}\right)$ $40 = 23 + 67e^{-kt}$ $17 = 67e^{-kt}$ $\left(\frac{17}{67}\right) = e^{-kt}$ $t = \frac{\ln\left(\frac{17}{67}\right)}{-k} \quad \checkmark$ $t = 38.68$ $t = 39 \text{ mins} \quad \checkmark$	2

6(c)	If $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$ prove $\sum_{r=1}^n r {}^nC_r = n.2^{n-1}$ $(1+x)^n = {}^nC_0(1)^n(x)^0 + {}^nC_1(1)^{n-1}(x)^1 + {}^nC_2(1)^{n-2}(x)^2 + \dots + {}^nC_n(1)^0(x)^n$ ☒ <i>Differentiate both sides :</i> $n(1+x)^{n-1} = {}^nC_1(1)^{n-1}(x)^0 + 2 {}^nC_2(1)^{n-2}(x) + \dots + n {}^nC_n(1)^0(x)^{n-1}$ ☒ If $x = 1$ $n(2)^{n-1} = {}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + \dots + n {}^nC_n$ ☒ $n(2)^{n-1} = \sum_{r=1}^n r {}^nC_r$	3
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Question	Criteria	Marks
7(a)(i)	$x = Vt \cos \theta$ $y = -\frac{1}{2}gt^2 + Vt \sin \theta$ $t = \frac{x}{V \cos \theta}$ $\therefore y = -\frac{1}{2}g\left(\frac{x}{V \cos \theta}\right)^2 + V\left(\frac{x}{V \cos \theta}\right)\sin \theta$ ☒ sub of t $y = \frac{-gx^2}{2V^2 \cos^2 \theta} + x \tan \theta$ ☒ rearranging and substitution $y = \frac{-gx^2}{2V^2} \sec^2 \theta + x \tan \theta$ $y = x \tan \theta - \frac{x^2}{4h}(1 + \tan^2 \theta)$ ☒ substitution of h and ($\sec^2 \theta = 1 + \tan^2 \theta$)	3
7(a)(ii)	$y = x \tan \theta - \frac{x^2}{4h}(1 + \tan^2 \theta)$ $X \tan \theta - \frac{X^2}{4h}(\tan^2 \theta + 1) - y = 0$ $X^2 \tan^2 \theta - 4hX \tan \theta + X^2 + 4hY = 0$ ☒ Hits at two different angles θ_1 and θ_2 when $\Delta > 0$ ☒ $\Delta = (4hX)^2 - 4(X^2)(X^2 + 4hY) > 0$ $16h^2X^2 - 4X^4 - 16X^2hY > 0$ $4X^4h < 16h^2X^2 - 16X^2hY$ $X^2 < 4h^2 - 4hY$ $X^2 < 4h(h - Y)$ ☒	3

7(b)(i)

1



$$\tan(\angle LEG) = \frac{a}{x} \Rightarrow \tan^{-1}\left(\frac{a}{x}\right) = \angle LEG$$

$$\tan(\angle MEG) = \frac{b}{x} \Rightarrow \tan^{-1}\left(\frac{b}{x}\right) = \angle MEG$$

since $\theta = \angle LEG - \angle MEG$

$$\therefore \theta = \tan^{-1}\left(\frac{a}{x}\right) - \tan^{-1}\left(\frac{b}{x}\right)$$



7(b)(ii)

3

$$\theta = \tan^{-1}\left(\frac{a}{x}\right) - \tan^{-1}\left(\frac{b}{x}\right)$$

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{a}{x}\right)^2} \times \frac{-a}{x^2} - \frac{1}{1 + \left(\frac{b}{x}\right)^2} \times \frac{-b}{x^2}$$



differentiation

$$\frac{d\theta}{dx} = \frac{-a}{x^2 + a^2} - \frac{-b}{x^2 + b^2}$$

$$\frac{d\theta}{dx} = \frac{-ax^2 - ab^2 + bx^2 + ba^2}{(x^2 + a^2)(x^2 + b^2)}$$



common denominator

$$\frac{d\theta}{dx} = \frac{-x^2(a-b) + ab(a-b)}{(x^2 + a^2)(x^2 + b^2)}$$

$$\frac{d\theta}{dx} = \frac{(a-b)(-x^2 + ab)}{(x^2 + a^2)(x^2 + b^2)}$$



grouping method

7(b)(iii)	$\frac{d\theta}{dx} = \frac{(a-b)(-x^2+ab)}{(x^2+a^2)(x^2+b^2)}$ $\frac{d\theta}{dx} = \frac{(3b-b)(-x^2+(3b)b)}{(x^2+(3b)^2)(x^2+b^2)} \quad \text{if } a = 3b$ $\frac{d\theta}{dx} = \frac{2b(-x^2+3b^2)}{(x^2+9b^2)(x^2+b^2)}$ max occurs when $\frac{d\theta}{dx} = 0$ $\therefore \frac{2b(-x^2+3b^2)}{(x^2+9b^2)(x^2+b^2)} = 0$ $2b(-x^2+3b^2) = 0$ $-x^2+3b^2 = 0$ $x^2 = 3b^2$ $x = \pm\sqrt{3}b \quad \boxed{\checkmark}$ since x is a distance then $x > 0$ then $x = \sqrt{3}b$ $\theta = \tan^{-1}\left(\frac{3b}{x}\right) - \tan^{-1}\left(\frac{b}{x}\right) \quad \text{if } a = 3b$ $\theta = \tan^{-1}\left(\frac{3b}{\sqrt{3}b}\right) - \tan^{-1}\left(\frac{b}{\sqrt{3}b}\right)$ $\theta = \tan^{-1}\left(\frac{3b}{\sqrt{3}b}\right) - \tan^{-1}\left(\frac{b}{\sqrt{3}b}\right)$ $\theta = 60^\circ - 30^\circ$ $\theta = 30^\circ \quad \boxed{\checkmark}$	
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